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DCBAB
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3.1 production function

The following questions refer to the diagram below, which shows the short-run production function of a perfectly competitive firm that produces potatoes using one variable input, labor.

Number of Workers	Potatoes Harvested (pounds per hour)
0	0
1	3
2	7
3	10
4	12
5	13
6	12

1. After which worker does diminishing marginal product first occur?

(A) Second worker ✓

(B) Third worker

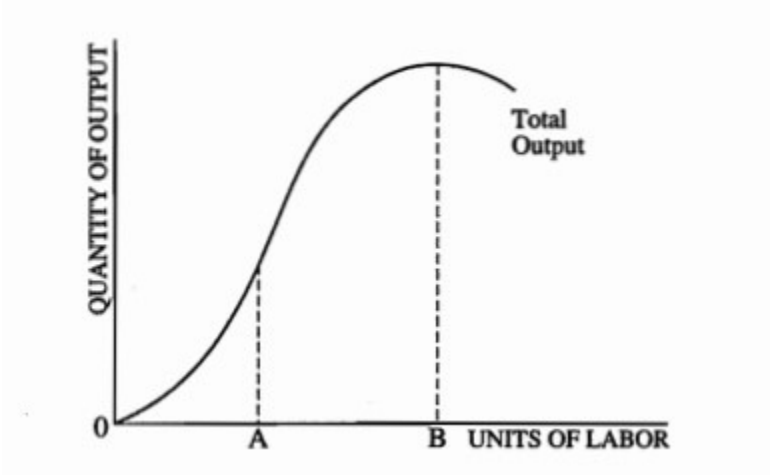
(C) Fourth worker

(D) Fifth worker

(E) Sixth worker

3.1 production function

2.



The relationship in the graph above best illustrates the economic concept of

(A) opportunity cost

(B) diminishing marginal utility in consumption

(C) diminishing marginal returns in production ✓

(D) production possibilities

(E) comparative advantage

3. Which of the following is true for a firm that uses labor as a variable input and capital as a fixed input in the short run?

(A) If the marginal product of labor is negative, the average product of labor must also be negative.

(B) If the marginal product of labor is rising, the average product of labor must be greater than the marginal product of labor.

(C) If the average product of labor is rising, the marginal product of labor must be rising.

(D) If the average product of labor is falling, the marginal product of labor must be less than the average product of labor. ✓

(E) The average product of labor can never be equal to the marginal product of labor.

Answer D

Correct. When the average product of labor is rising, the marginal product is above it, pulling the average up. When the marginal is below the average product, the average product is falling because the marginal product is pulling it down. The average product is at its maximum when the marginal product equals the average product. Therefore, if the average product of labor is falling, the marginal product of labor must be less than the average product of labor.

3.1 production function

4. As its output increases, a firm's short-run marginal cost will eventually increase because of

(A) diseconomies of scale

(B) a lower product price

(C) inefficient production

(D) the firm's need to break even

(E) diminishing returns ✓

5. Assume that labor is the only variable input. If a firm's short-run marginal cost is increasing as output rises, which of the following must be true?

(A) Average product of labor is constant.

(B) Average product of labor is increasing.

(C) Marginal product of labor is greater than average product of labor.

(D) Marginal product of labor is decreasing. ✓

(E) Total product of labor is decreasing.

6.

Number of Workers	Bushels of Apples Picked
0	0
1	4
2	9
3	15
4	20
5	24

The table above shows the short-run production function for picking apples. Based on the production data, which of the following statements about the marginal product of the fifth worker is true?

(A) It is the maximum that can be attained.

(B) It is greater than the marginal product of the first worker due to increasing returns.

(C) It is greater than the combined marginal products of all the other workers.

(D) It is less than the marginal product of the third worker due to diminishing returns. ✓

(E) It is rising due to increasing marginal returns.

3.1 production function

7. The table below shows a production function for a firm.

Units of Variable Input	Total Product
1	10
2	22
3	40
4	60
5	68
6	74
7	76
8	68
9	50
10	20

All of the following can be concluded from the information in the table EXCEPT:

(A) Diminishing marginal returns set in after the fourth unit of the variable input.

(B) Marginal product is positive if total product is increasing.

(C) This is a production function for a perfectly competitive firm. ✓

(D) A profit-maximizing firm would never voluntarily employ the eighth unit of the variable input.

(E) Total product starts to decrease when marginal product changes from positive to negative.

Number of Workers	Quantity of Output
0	0
1	10
2	25
3	35
4	40
5	42

8. Given the production schedule, what is the maximum number of workers the firm can hire before the effects of diminishing marginal returns set in?

(A) 1

(B) 2 ✓

(C) 3

(D) 4

(E) 5

3.1 production function

Answer B

Correct. The marginal product of the first worker is 10, of the second worker is 15, of the third worker is 10, and so on. The marginal product begins to decline with the third worker. The maximum number of workers the firm can hire before diminishing returns set in is 2.

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9. With capital held constant, when Zane employs one additional worker, the marginal product of labor decreases by 5 units. Which economic concept best characterizes Zane's production process?

(A) Diminishing marginal returns ✓
(B) Increasing marginal returns
(C) Constant returns to scale
(D) Economies of scale
(E) Diseconomies of scale

Answer A

Correct. As a firm employs more workers, each additional worker will have less capital to work with, and therefore the firm will experience diminishing marginal returns for each additional worker hired.

10. Suppose that a firm begins to hire workers for a newly completed plant with a fixed amount of machinery. As the firm hires additional workers, one would expect the marginal product to

(A) fall initially, but eventually rise
(B) rise initially, but eventually fall ✓
(C) rise consistently due to diminishing return
(D) rise consistently due to the advantages of specialization
(E) rise consistently due to economies of scale

3.1 production function

The following question is based on the output and cost data in the table below.

Quantity of Output Produced (units)	Total Variable Cost ($)	Total Cost ($)
1	100	200
2	190	290
3	270	370
4	340	440
5	420	520
6	510	610
7	610	710

11. If there is only one variable input, diminishing marginal returns first occur with the production of which unit of output?

(A) 7th

(B) 6th

(C) 5th ✓

(D) 4th

(E) 3rd

12. If labor is the only variable input in the production process, the short-run marginal cost curve is upward sloping because which of the following occurs as more and more labor is added?

(A) Output decreases, and thus marginal cost increases.

(B) Output increases, and thus marginal cost increases.

(C) Output increases at an increasing rate, and thus the cost of producing each additional unit of output increases.

(D) Output increases at a decreasing rate, and thus the cost of producing each additional unit of output increases. ✓

(E) Output increases at a decreasing rate, and thus the cost of producing each additional unit of output decreases.

13. In microeconomics, the short run is defined as which of the following?

3.1 production function

(A) A period that is less than one year

(B) A period that is between one year and four years

(C) A period that is too short for a firm to be able to change its level of output

(D) A period during which some inputs in a firm's production process cannot be changed ✓

(E) A period during which a firm's fixed costs exceed its variable costs

14. Which of the following provides an example of the law of diminishing returns?

(A) A farm doubles all its inputs and observes that output less than doubles.

(B) A textile producer's output increases by 5 units when adding a fourth worker and by 7 units when adding a fifth worker.

(C) If a fixed input increases, fewer additional units of a variable input will be needed to increase output by one unit.

(D) As more of a variable input—for example, labor is used with a fixed number of machines— output increases but at a diminishing rate. ✓

(E) More cooks in a single kitchen will increase the number of meals produced by increasing amounts.

Answer D

Correct. The law of diminishing returns states that as more of a variable input is added to a fixed input, output eventually increases at a decreasing rate. This is a short-run concept using one variable input and at least one fixed input.

15. Locotek produces toy trains and pays each worker $350 per week. Five workers can produce 40 trains per week and six workers can produce 45 trains per week. The marginal product per week of the sixth worker is

(A) $70

(B) $350

(C) 5 trains ✓

(D) 7.5 trains

(E) 42.5 trains

16. If the marginal product of an input is positive and increasing, total product is

(A) decreasing at an increasing rate

(B) increasing at an increasing rate ✓

(C) increasing at a decreasing rate

(D) decreasing at a decreasing rate

(E) increasing at a constant rate

3.1 production function

Answer B

Correct. Marginal product is the change in total product as a result of increasing the quantity of an input (such as labor) by one unit. If marginal product is positive, then total product will increase by the amount of the marginal product. If marginal product is increasing, then total product will also increase at that increasing rate.

17.

Number of Workers	Total Product (units)
0	0
1	11
2	20
3	27
4	32
5	35
6	36

The table above shows a firm's short-run production function using labor as the only variable input. The marginal product of the third worker is

(A) 5 units

(B) 7 units ✓

(C) 9 units

(D) 20 units

(E) 27 units

Answer B

Correct. Marginal product is the change in total product that occurs as a result of an additional unit of labor. As the table indicates, output increases from 20 to 27 as a result of adding the third worker. Therefore, the marginal product for the third worker is 7 units.

3.1 production function

The following two questions refer to the data in the table below.

A perfectly competitive firm operates with a fixed amount of capital that costs $\$1{,}000$ per day. Labor is the only variable input. The firm hires labor in a perfectly competitive labor market at $\$100$ per day per worker. The table below shows the firm's production function.

Number of Workers Hired	Quantity of Output (units)
0	0
1	10
2	30
3	54
4	75
5	85
6	90

18. What is the marginal product of the third worker?

(A) 5
(B) 10
(C) 20
(D) 24 ✓
(E) It cannot be determined from the information given.

Answer D

Correct. The marginal product of the third worker is 24 units. The quantity of output with two workers is 30 units; the output increases to 54 when the third worker is added. Therefore, the marginal product of the third worker is $(54 - 30)/(3 - 2) = 24$ units.

19. Assume the marginal product of labor first rises, reaches a maximum, and then falls. If the average product of labor is falling, which of the following is true?

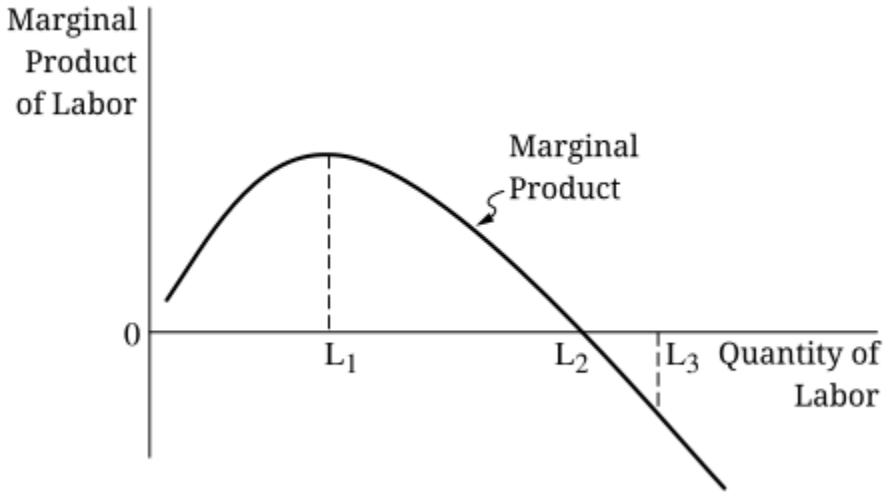
3.1 production function

(A) The marginal product of labor is greater than the average product of labor.
(B) The level of output produced must be at its maximum.
(C) The marginal product of labor must be falling. ✓
(D) The marginal product of labor must be negative.
(E) The average product of labor has not yet reached its maximum.

Answer C

Correct. The average product of labor will also rise, reach a maximum, and then decrease. If the average product of labor is decreasing, the marginal product of labor is falling and is below the average product of labor, reducing the average product of labor.

20.



Which of the following will happen if the firm continues to hire additional units of labor beyond L_2 in the short run?

(A) The firm's marginal cost will decrease.
(B) The firm will begin experiencing diminishing marginal returns.
(C) The firm's fixed cost will increase.
(D) The average product of labor will be equal to zero.
(E) The total product of labor will decrease. ✓

3.1 production function

Answer E

Correct. At quantities of labor hired beyond L_2, the marginal product of labor is negative. When the marginal product of labor is negative, additional labor hired will decrease total product.

21. A firm's short-run production function uses capital and labor as inputs. Assume the quantity of capital is fixed in the short run. The marginal product of labor will decrease as more labor is used in the short run because each unit of labor will have

(A) zero units of capital to work with

(B) fewer units of capital to work with ✓

(C) an increase in the capital-to-labor ratio

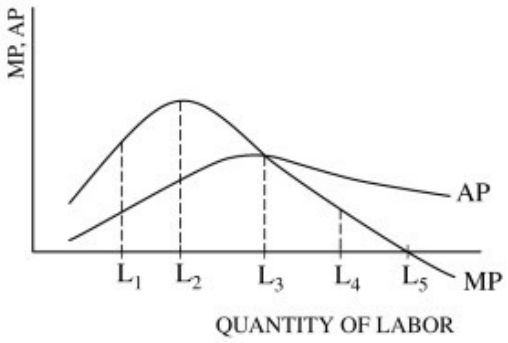
(D) more units of capital to work with

(E) an infinite number of units of capital to work with

Answer B

Correct. In the short run, as the firm hires more labor to produce more output, each additional unit of labor will have to share the fixed capital that the firm has; therefore, there will be a decrease in the marginal product of labor. Each unit of labor will have fewer units of capital to work with.

22.



The graph above shows the marginal product (MP) and the average product (AP) of labor for a firm that uses labor as the only variable input and hires its labor in a perfectly competitive market. At which quantity of labor does marginal cost change from decreasing to increasing?

3.1 production function

(A) L_1

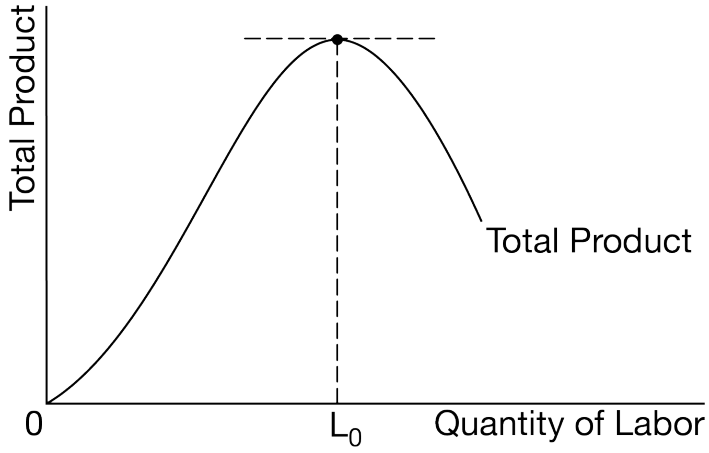
(B) L_2 ✓

(C) L_3

(D) L_4

(E) L_5

23.



Based on the short-run production function graph above showing the relationship between the quantity of labor and total product, which of the following statements is true?

(A) The marginal product of labor is always increasing.

(B) At L_0 the marginal product of labor is at its maximum.

(C) Total product is maximized when marginal product is zero. ✓

(D) At L_0 the marginal product of labor exceeds the average product of labor.

(E) The marginal product of labor is always positive.

Answer C

Correct. With this short-run production function, the marginal product of labor initially rises, then begins to fall. At L_0 the marginal product of labor is zero and total product is at its maximum because the change in total product is zero. When the change in total product is zero, adding an additional unit of the variable input results in a decline in total product.

24. Which of the following is true of production in the long run?

3.1 production function

(A) At least one factor of production is fixed.
(B) Only one factor of production is fixed.
(C) Some factors of production are fixed while others are variable.
(D) All factors of production are variable. ✓
(E) All factors of production, except technology, are fixed.

Answer D

Correct. In the long run, firms have complete flexibility in utilizing all factors of production. Therefore, all factors of production are variable in the long run.

25. A short-run production function shows the relationship between a firm's inputs and the output it produces when the firm

(A) does not vary any of its inputs
(B) has at least one fixed input ✓
(C) varies all of its inputs
(D) has constant marginal cost
(E) has no fixed costs

Answer B

Correct. In the short run, the firm employs fixed and variable inputs. There needs to be at least one fixed cost in the short run. The fixed input could be the number of machines or the amount of farmland.

26. Given a short-run production function, which of the following is true when total product is increasing at a decreasing rate?

(A) Average product must be positive and increasing.
(B) Average product must be negative and decreasing.
(C) Marginal product must be negative and decreasing.
(D) Marginal product must be positive and increasing.
(E) Marginal product must be positive and decreasing. ✓

27. Which of the following is true when total product is at its maximum?

3.1 production function

(A) Marginal product is at its maximum.

(B) Marginal product is equal to zero. ✓

(C) Marginal product is equal to average product.

(D) Average product is increasing.

(E) Average product is equal to zero.

Answer B

Correct. When marginal product equals zero, total product is maximized because the change in total product is zero. Because marginal product is diminishing, adding additional units of variable input beyond this point will result in negative marginal product and a decline in total product.

28. When total physical product is at its maximum, marginal physical product must be

(A) greater than one

(B) equal to one

(C) equal to zero ✓

(D) less than zero

(E) constant

29. When marginal product exceeds average product, which of the following must be true?

(A) Average product is increasing. ✓

(B) Average product is decreasing.

(C) Marginal product is increasing.

(D) Total product is decreasing.

(E) Total product is at its maximum.